Trustworthy Collective Reasoning and Evaluation for AI Agents

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1 The problem

Five AI agents read a set of shipping reports and are asked whether a cargo vessel has left port. Three answer yes and two answer no. The vessel is in fact still in port. Two of the yes answers rest on the same outdated schedule, and the third comes from an agent that has been compromised and argues confidently for yes. Nobody knows in advance which agents are reliable on questions like this. A majority vote picks the wrong answer, and an open discussion may do no better, because the compromised agent is persuasive and the two agents with the same source reinforce each other.

The central problem of this project is how a group of AI agents of unknown and correlated reliability, some possibly dishonest, can reach trustworthy decisions when only a few claims can be checked.

Collective judgment fails in three ways, all present in the example. Counting every vote equally lets less reliable agents outvote experts, and agents that make the same mistakes count more than once. Agents may behave strategically: they can rate their own answers too highly, rate rivals low to raise their own standing, argue for an answer they do not hold, or coordinate with one another. In discussion, agents give way to whoever is most confident or persistent, with or without evidence. Three research goals follow: how much weight each agent deserves (RG1), which claims to check and how to settle disputes (RG2), and how verified outcomes should improve the weights and the agents (RG3). We focus on factual claims and reasoning steps that can be checked, at least in part.

2 Why it matters

Using one model to judge another is now standard practice in AI evaluation and oversight [1, 2], and multi-agent systems increasingly decide by vote or discussion. Agreement among these models is weak evidence, because their errors are strongly correlated [3], and more capable models make more similar mistakes [4]. In one measurement, nine judges were worth about 2.2 independent judges, items on which all nine agreed were still wrong 9.1% of the time, and established aggregation methods closed at most 11% of the gap to an independent panel, even with access to the correct answers [5]. Discussion erodes independence further: one round of debate raised a panel's unanimity from 39.5% to 95.2% while accuracy moved less than a point [6]. When breaking a checking protocol paid more than following it, two agents repeatedly checking each other's work colluded in 94% of runs across ten models, with stronger models in a family colluding sooner [7].

Naval settings raise each of these risks. AI tools differ in reliability across tasks, and that reliability is rarely known in advance. Ground truth arrives late and sparsely. An adversary may compromise an agent or inject false reports, and checks can be costly or impossible under limited connectivity, yet a commander or analyst must still decide whether to act on what the AI agents agree on.

3 How it is solved today, and the limitations

Statistical methods for crowdsourcing estimate each rater's accuracy without an answer key, jointly with item difficulty [8, 9], and truth-discovery methods detect sources that copy one another [10]. They assume that raters err independently or that their dependence has a simple structure [11], and they treat each judgment as fixed. Two recent preprints use small trusted sets to study language-model judges as honest signals that never interact: one estimates accuracy and shared mistakes to choose a voting rule [12], and the other drops redundant judges and routes questions to specialists within slices declared in advance [13].

Computational social choice offers mechanisms in which a participant's ratings of others cannot change its own chance of selection [14, 15], along with fair, manipulation-resistant panel selection [16, 17] and robust ranking of AI models [18]. Lev et al. showed that an impartial mechanism can weight reviewers by reliability [19]; their weights come from agreement with co-reviewers, which rewards siding with the majority, and the authors note that the true ranking is unrecoverable once accurate reviewers are half or fewer. These

mechanisms combine preferences or select impartially, taking inputs as given. Here there is a correct answer, reliability is unequal and correlated, judgments change during discussion, and some claims can be checked.

Multi-agent debate lets models critique one another [20], but majority voting accounts for most of its gains [21], and a single persuasive adversary can cut a debating group's accuracy by roughly 10% to 40% [22]. In AI safety debate, two models argue before a judge [23], and restricting them to verified quotations helps the judge reach the truth [24]. In a recent collaborative variant, two disagreeing models locate the crux of their disagreement before a weak judge decides. Weak-judge accuracy was 62.1%, against 49.2% under adversarial debate, but the judges accepted the models' consensus almost every time, including when both models were wrong, and the method assumes two honest models [25]. Colluding agents can also mislead judges when every fragment they cite is true [26].

Peer prediction rewards an agent for reports that help predict other agents' reports, which can make honest reporting an equilibrium without ground truth; a recent extension to language models resists a single deceiver and leaves collusion open [27]. AI control tests oversight protocols against a red team that tries to subvert them, using a trusted weaker model and a small audit budget [28]. None of this work learns whom to trust from verified outcomes while protecting the group's decision from persuasion and coordinated dishonesty.

4 Our approach: what is new, and why it will succeed

We anchor every step in a small stream of verified checks. Every claim has some chance of being checked, decided by random draws that agents cannot predict. Each agent first answers on its own, splits its reasoning into checkable claims, and cites its sources; this sealed first answer is kept for later comparison. The goal is accurate, calibrated decisions within a checking budget that hold up when some agents are dishonest and that mark a claim unresolved when the evidence runs out.

What is new. (i) Weights come from a model of per-question-type skill and shared failures, learned from randomized audits that strategic agents cannot anticipate. (ii) Skill-weighted aggregation stays impartial, and we measure what coalitions trading ratings across groups can still gain. (iii) Checks are allocated so a coalition knowing the policy cannot tell which claims are safe to corrupt. (iv) Disputes among many correlated agents are settled only by verifiable evidence, and agents are rewarded for the verified evidence they add.

Why it will succeed. Two results make the approach feasible. Without trusted checks, adversaries that imitate reliable raters cannot be told apart from them, and trusted effort that does not grow with the number of items suffices to separate them even when reliable raters are a minority [29]. Spot checks against occasional trusted evaluations also give stronger incentives for honest effort than many peer-prediction mechanisms while using less ground truth [30]. PRD, from the PI's group, introduced peer ranking and peer discussion among language-model evaluators and agreed better with human judgments than the strongest single model used as judge, which favors its own answers [31]; our PeerEval leaderboard (peereval-ai.github.io) applies peer evaluation to many current models. FG-PRM detects six types of reasoning errors at the level of individual steps [32], and HiPRAG uses step-level rewards to train search agents to search only when needed, reducing unnecessary searches to 2.3% [33]. The PI also co-led one of five red teams in the Amazon Nova AI Challenge (teams were selected from more than 90 proposals), probing code-generating models for flaws. The PI holds a 2024 NSF CAREER award and a 2023 Amazon Research Award on process-guided fine-tuning, and as a postdoctoral researcher led system integration for RESIN-11 in DARPA KAIROS.

RG1. Learn whom to trust

Task 1.1: Expertise- and dependence-aware weighting. We will estimate each agent's skill on each type of question and each question's difficulty, extending the model of Dawid and Skene [8, 9] with shared error factors. Two agents share an error factor when they fail together on audited claims more often than their separate accuracies predict. Shared sources, the same underlying model, and overlapping reasoning steps help predict such factors [10, 4], but agents that share a source can still reason differently, so the estimate comes from coinciding failures. Unlike the declared slices of Zhu et al. [13] or the fixed rules of Hossain et al. [12], we learn question types and difficulty. Because each check is weighted by its known selection

probability (Task 3.1), the estimates stay unbiased even for an agent that behaves well only where checks are likely, and we will measure how many audits this requires. In the vessel example, the two agents relying on the same schedule fail together on audited claims, so their agreement counts for little more than one vote.

Task 1.2: Impartial aggregation that resists coalitions. When agents rate one another, as on our PeerEval leaderboard, weighting by skill can conflict with impartiality. In PRD's ranking, an agent that lowers a rival's score also lowers that rival's weight as a judge, which can raise its own score. We will derive each agent's weight as a judge only from audits that no agent's reports can influence: a uniform subsample of the random checks and audits from earlier rounds. Following the bipartite assignment of Lev et al. [19] and partition mechanisms [15], groups are fixed before any ratings, and ratings flow only between groups; agents built on the same underlying model share a group, since a model tends to recognize and favor its own outputs [34]. Partitioning stops agents in one group from boosting each other, so the remaining threat is agents in different groups trading favorable ratings. We will measure how much such a coalition of k agents can gain and flag groups whose agreement exceeds what their measured dependence explains. For selecting top candidates, we will adapt randomized impartial selection [14], using RG1's skill estimates as predictions [35], and report the accuracy that impartiality costs.

RG2. Decide what to check, and settle disputes with evidence

Both tasks follow one rule: *a challenge can change the group's answer only through verifiable evidence*, meaning a record or test result obtained by a check, a quotation verified against its source, or a reasoning step simple enough to verify directly.

Task 2.1: Decide what to check. Checking every claim is too costly, and checking only where agents disagree misses the most dangerous errors, because correlated agents agree on the same wrong answer and colluding agents can simply agree. We will treat the choice of which claims to check as a game against an adversary that knows our policy but not its random draws. Each batch of checks mixes four streams: a random sample that gives every claim a minimum chance of being checked and extra weight to claims that could change a decision, which keeps estimates unbiased and leaves no claim safe; claims where the weighted disagreement could change the decision, found by aligning the agents' claims and ranking suspect steps with FG-PRM [32]; claims on which agents that RG1 estimates to be dependent agree more than their shared evidence explains; and claims flagged by the dishonesty signals of Task 2.2. We will bound how many wrong decisions a coalition of k agents can cause before it is detected, including a coalition that behaves well whenever it expects a check, and derive how to split a budget across the four streams. In the vessel example, one check of the latest port-authority entry settles the dispute, and random checks of earlier claims can catch the compromised agent where it did not expect a check. Baselines check only disagreements, only the least confident claims, or uniformly random claims, at matched budgets.

Task 2.2: Debate among many agents, settled by evidence. For disputes that no single check can settle, because the solutions differ too much to align or no record or tool is reachable, we will extend debate to many agents of unknown and correlated reliability, comparing adversarial debate before a judge [23, 24] with collaborative disagreement resolution, in which agents locate the crux of their disagreement and revise [25]; once a dispute narrows to a checkable claim [23], Task 2.1 checks it. Speakers are drawn at random from each side's most reliable agents, as in random panel selection [16], and agents whose errors are strongly correlated share one speaker, so repeating a view adds no weight [36]. Only verified quotations and directly verifiable steps count as evidence, and the opposing side may add missing context, because true fragments can still mislead [26]. Judges are drawn at random from agents held out of the discussion that share no underlying model with either side [4], weighted by their audited record as judges. We will track three signs of dishonesty (arguing against one's own sealed first answer without new evidence, citing evidence that does not support the claim, and changing others' answers without evidence) and feed them into RG1 and Task 2.1. The evaluation measures how often a wrong answer is corrected, how often a correct one is lost, and how many dishonest agents each format tolerates.

RG3. Learn from verified outcomes

Task 3.1: Update trust without selection bias. Each dispute settled by a check becomes a new audited

claim that updates RG1. Targeted checks over-represent errors, so each check is weighted by the inverse of its known selection probability; because answers are sealed before the draw, this keeps the estimates unbiased [37]. Outcomes decided only by debate update the weights provisionally and never become audits, so persuasive errors cannot reinforce themselves. Trust is tracked per question type and decays over time, which limits how much an agent gains by behaving well early and defecting later.

Task 3.2: Reward verified contributions. Training debaters to win can improve judge accuracy [38], but even debate-trained critics have fabricated quotations against a weakened judge [39], and training on human approval has taught models to mislead human evaluators [40]. We will reward each message for its verified effect. A message earns reward when it introduces verified evidence that moves the group toward the correct answer. Revising in response to such evidence also earns reward, but less than a correct first answer, so starting out wrong never pays; changing others' answers without evidence, giving way without evidence, and abandoning a correct first answer are penalized. To decide which message moved the group, we will replay the discussion without it and score the reasoning inside each message with FG-PRM-style step rewards, building on multi-agent reinforcement learning for language models [41, 42]. Shared rewards between agents that check each other can produce collusion [7], so each agent is rewarded separately and training episodes are audited at random. The main experiment compares training to win with training to contribute, measuring accuracy, lost correct answers, fabricated evidence, and emergent collusion.

Task 4 (shared). Testbeds and a red-team evaluation

All scenarios are synthetic or drawn from public data. (A) Math and code problems solved by agents of known skill, some given the same flawed hint, so expertise and dependence are known by construction. (B) Multi-document report assessment built with our MEQA question-generation pipeline [43] and from MuSiQue [44], including synthetic maritime scenarios with outdated and conflicting reports. (C) Scientific claims and peer review, using public conference reviews and author rebuttals, with claims later confirmed or contradicted by reproducibility studies serving as audits. On these testbeds we will run a control evaluation [28]: a red team trains agents to collude, to coordinate through stylistic signals, and to attack only when they expect no check, while the blue team uses the methods above, trusted weaker models, canary cases with known answers, and limited audits, and we will extend a public two-agent collusion environment [7] to many agents. Results are curves of safety (wrong decisions avoided) against usefulness (correct decisions made without escalation) as the audit budget varies. Baselines include majority vote, Dawid-Skene weighting, PRD, and the closest recent methods [12, 13, 19, 27, 25], at matched compute and checking budgets.

5 Limitations of our approach

If every agent shares the same misconception and nothing can be checked, no aggregation or debate can recover the truth; our methods should report such claims as unresolved. The approach needs some trusted checks, and checks can themselves be wrong, so we will measure how sensitive each method is to unreliable checks and judges. A coalition that defects once, on a single high-stakes claim, may cause a wrong decision before detection; Task 2.1 bounds such losses without ruling them out. Robustness will be shown only against the dishonest and colluding behaviors we test, because agents can coordinate through hidden signals that monitoring and paraphrasing do not fully remove [45]. Operational naval use would require domain data and expert validation beyond this fundamental research.

6 Impact if successful

The project will show when agreement among AI agents counts as evidence and how many checks it takes to make it count. Each task targets a property of naval settings: unknown reliability (Tasks 1.1 and 3.1), late and sparse ground truth (Tasks 2.1 and 3.1), compromised agents (Tasks 1.2, 2.2, and 3.2), and costly or unavailable checks (Tasks 2.1 and 2.2). For an analyst using several AI tools, the result would show which tools' judgments carry weight and why, identify the claim still in dispute, and say when a decision needs human review. The same methods would let conferences and funding agencies test impartial, audited peer review with language-model reviewers. We will release code, testbeds, and evaluation protocols.

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